pfi

Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Randomized Search and Honest Tuning

WEEK 3 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Week 3: the Polish rows, and today's question

Course 4's Polish training rows

4,680

training rows

326

bankrupt (6.97%)

Every fit, score, and choice by cross-validation inside these rows. Course 4's 1,170 test rows stay closed.

Today's question

A search always finds a best candidate. Is it better, or only the luckiest?

Primary metric: average precision, because the outcome is rare.

Allowed names

model probability of bankruptcy in this file

flagged at threshold t

Never called

a credit view on any company

a rating

The default model, and what a grid costs

0.7783

default LightGBM, mean of 5 folds

0.0469

fold standard deviation

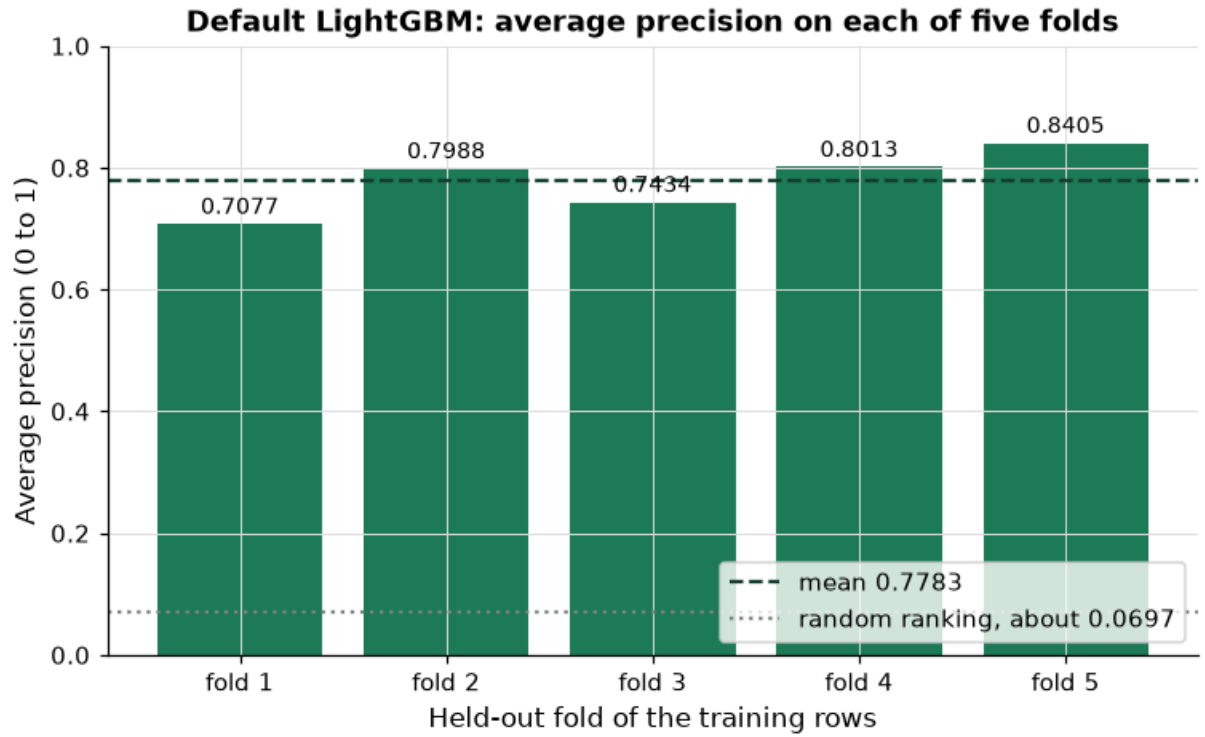
100 trees, 31 leaves, learning rate 0.1, at least 20 rows per leaf

Grid	Candidates	Models (5 folds)
2 values of 3 settings	8	40
5 values of 3 settings	125	625

Cost is candidates times folds, and each setting multiplies it.

A two-value grid only ever looks at two values of each setting.

One model, five fold scores



Same model, five scores

0.7077
fold 1

0.8405
fold 5

All five: 0.7077, 0.7988, 0.7434, 0.8013, 0.8405.
The spread is the rows, not the model.

Read every tuning gain today against bars like these.

A randomized search: fix the budget, draw the settings

```
space = {"num_leaves": randint(8, 64),
         "learning_rate": loguniform(0.02, 0.3),
         "min_child_samples": randint(5, 60)}
```

n_iter = 6, under the grid's 8

6 candidates: up to 6 values of every setting, where the grid tries 2. random_state=SEED fixes the draws.

The search's best

num_leaves 46, learning_rate 0.1452, min_child_samples 25: best_score_ 0.7760

loguniform: 0.02 to 0.03 is as likely as 0.2 to 0.3.

Reading the search: means inside each other's noise

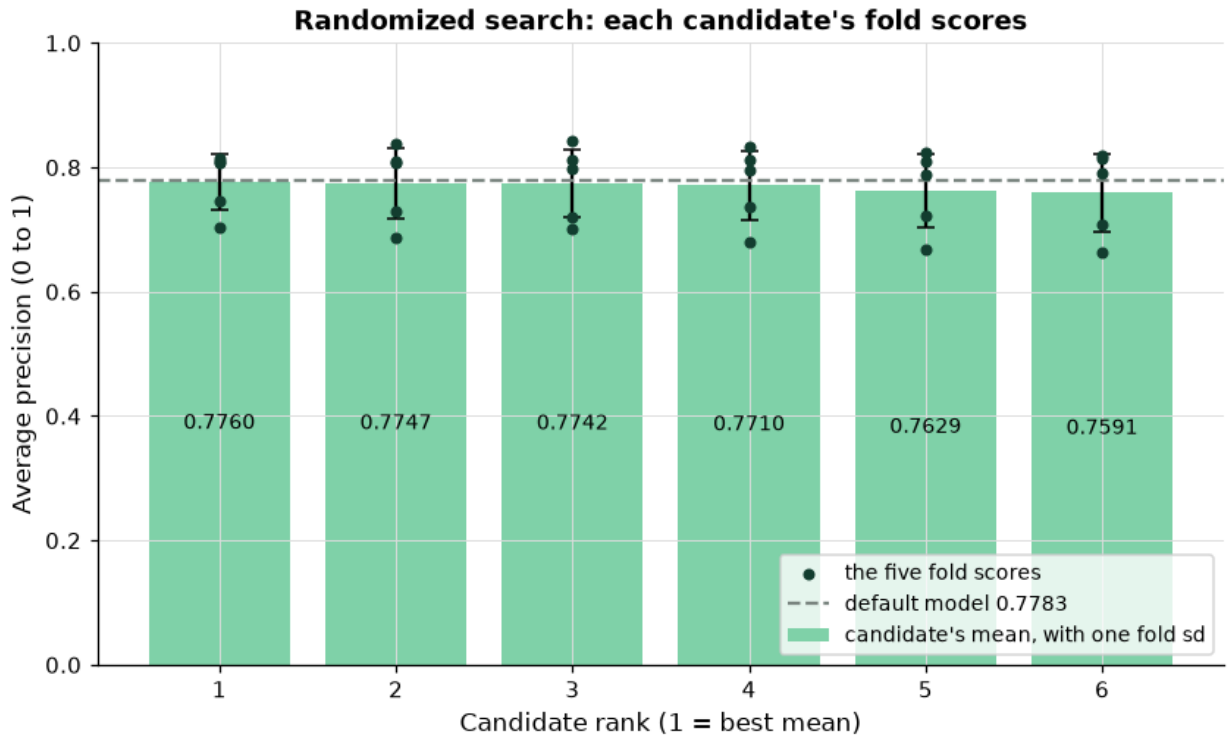
Rank	Mean	SD	Leaves	Rate	Rows/leaf
1	0.7760	0.0445	46	0.1452	25
2	0.7747	0.0564	22	0.0551	33
3	0.7742	0.0550	43	0.0694	57
4	0.7710	0.0561	18	0.0669	27
5	0.7629	0.0588	29	0.0294	7
6	0.7591	0.0622	51	0.0233	28

search_table

Every candidate's fold standard deviation is larger than the gap between any two neighbouring means.

42 tests pass after today's three functions.

Each candidate's fold scores



Folds against settings

0.0169

means apart

0.1130

least fold span

Fold 1 is the lowest dot of every candidate, and of the default model.

The folds make most of the spread; the settings a small part of it.

Tuning's gain against the folds' own noise

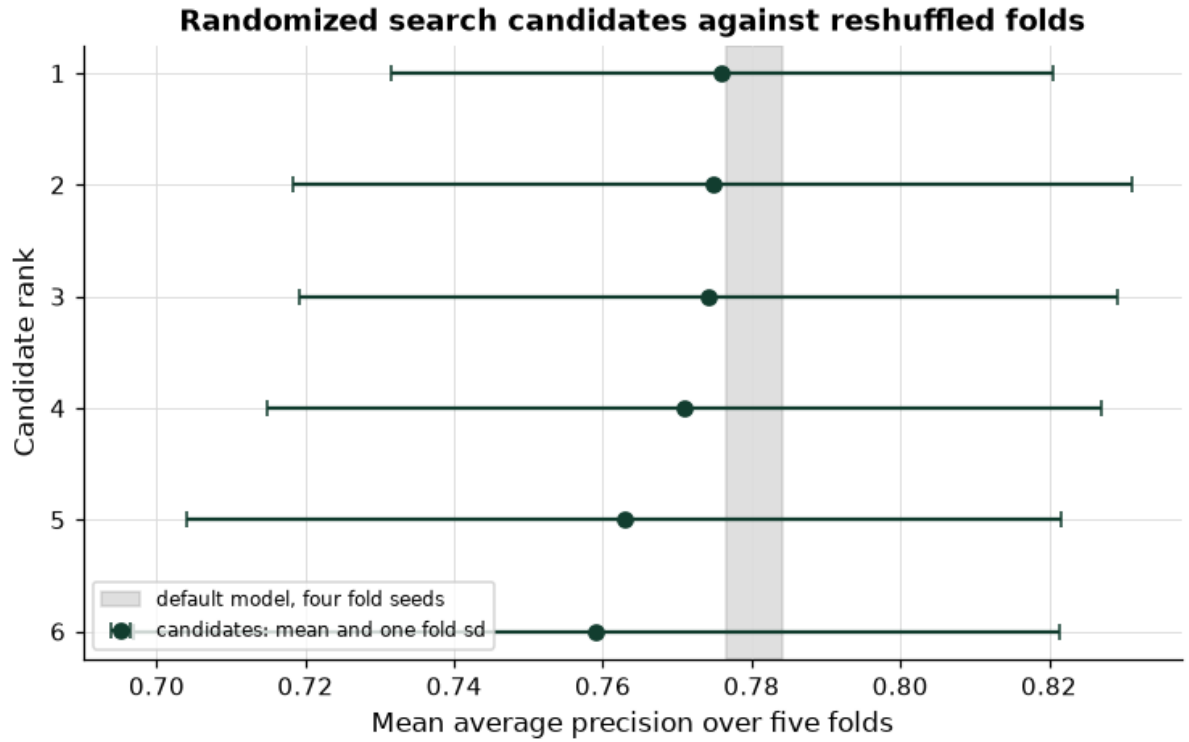
-0.0024

gain from tuning

0.0075

range, default on 4 fold seeds

Fold seed	Default mean
42	0.7783
43	0.7766
44	0.7841
45	0.7838



Tuning moves the mean less than reshuffling the folds does.

Nested cross-validation: an honest score

1. Hold out

one outer fold

2. Search

on the other four, inner folds

3. Refit, score

on the held-out fold

4. Repeat

for all five

0.7751

nested mean

0.7760

best_score_

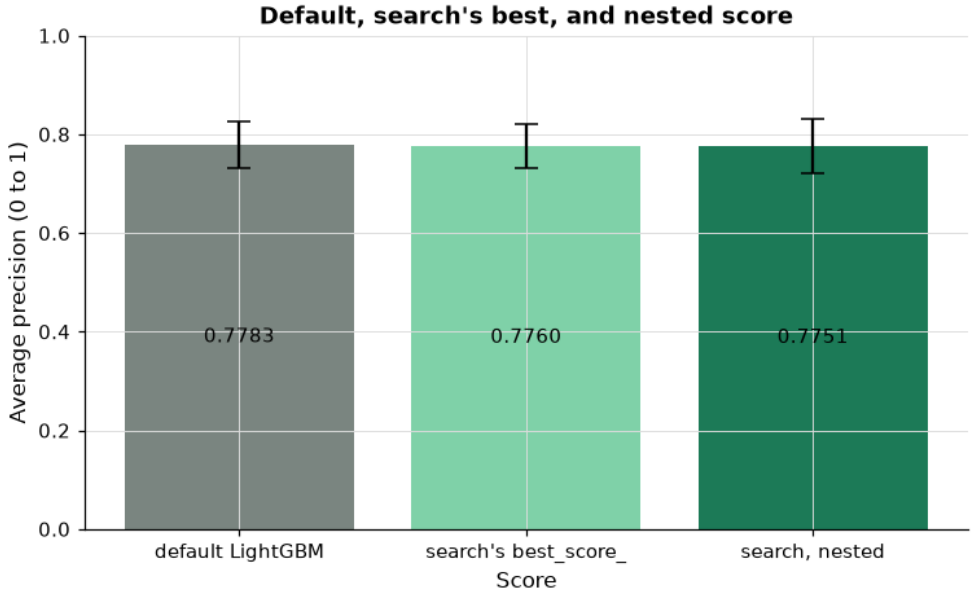
0.7783

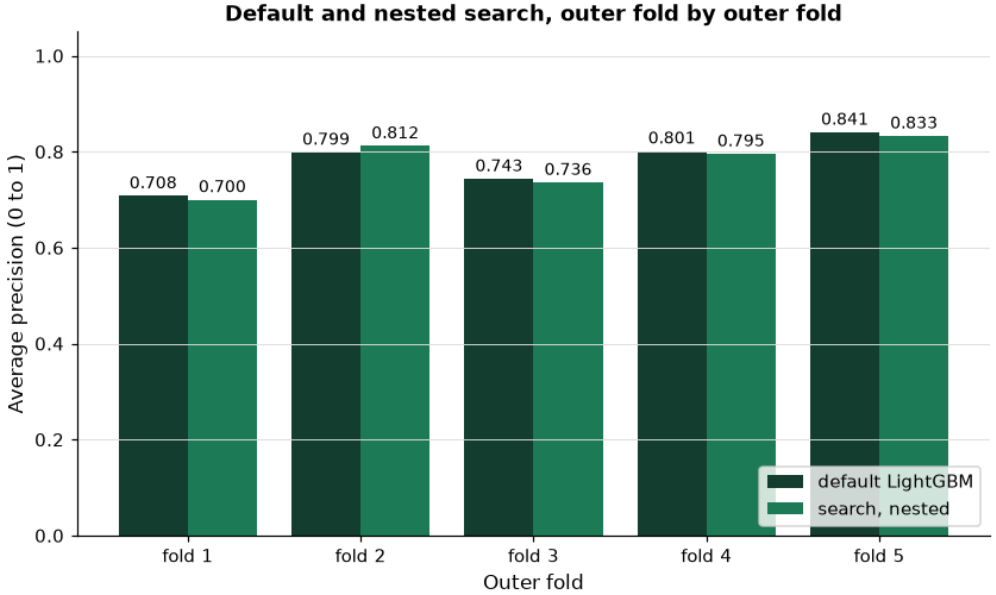
default model

Outer folds: 0.7000, 0.8120, 0.7358, 0.7953, 0.8327; sd 0.0554

Scored honestly, the tuned model does no better than the defaults on these rows.

Three scores, and the folds behind them





Score	Mean	Fold sd	Rows that chose the settings
default LightGBM	0.7783	0.0469	none chosen
search's best_score_	0.7760	0.0445	the same five folds
search, nested	0.7751	0.0554	inner folds only

Fold 1 hardest for both; nested higher on 1 of 5.

What to tune first, and when to stop

Defaults first	Score them: the baseline every search must beat
Tree size next	num_leaves or max_depth, rows per leaf
Then rate and trees	Together; early stopping picks the count
Fix the budget	n_iter and folds written before you look (n_iter=6)
Stop inside the noise	Gain below a fold sd or the reshuffle range

Practice that has served well, not a rule. Log every search with its nested score.

The threshold is a choice too

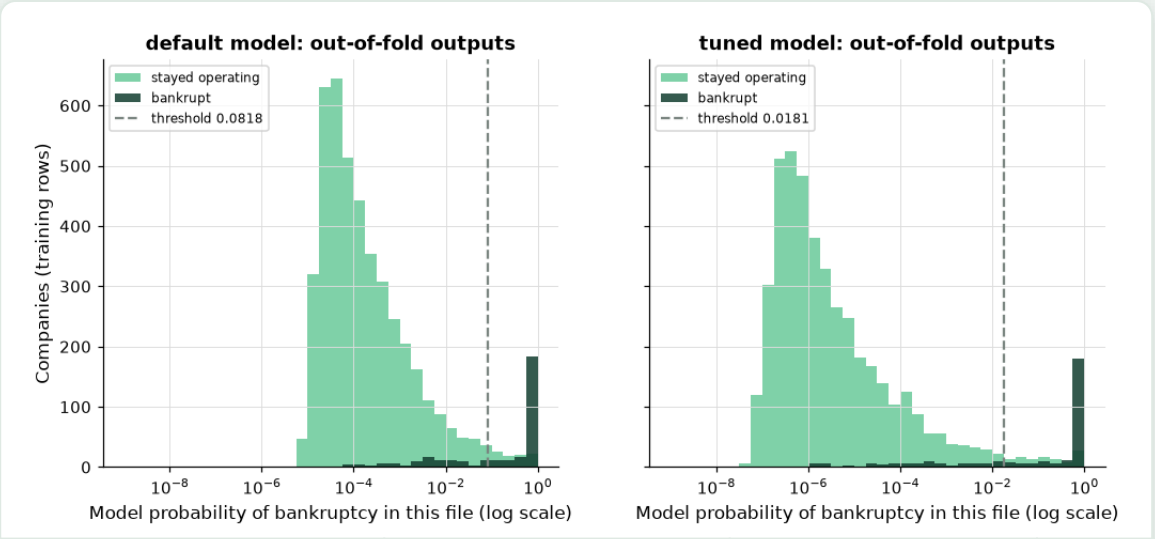
Model	Threshold	Flagged	Precision
Default	0.0818	327	0.7003
Tuned	0.0181	336	0.6815

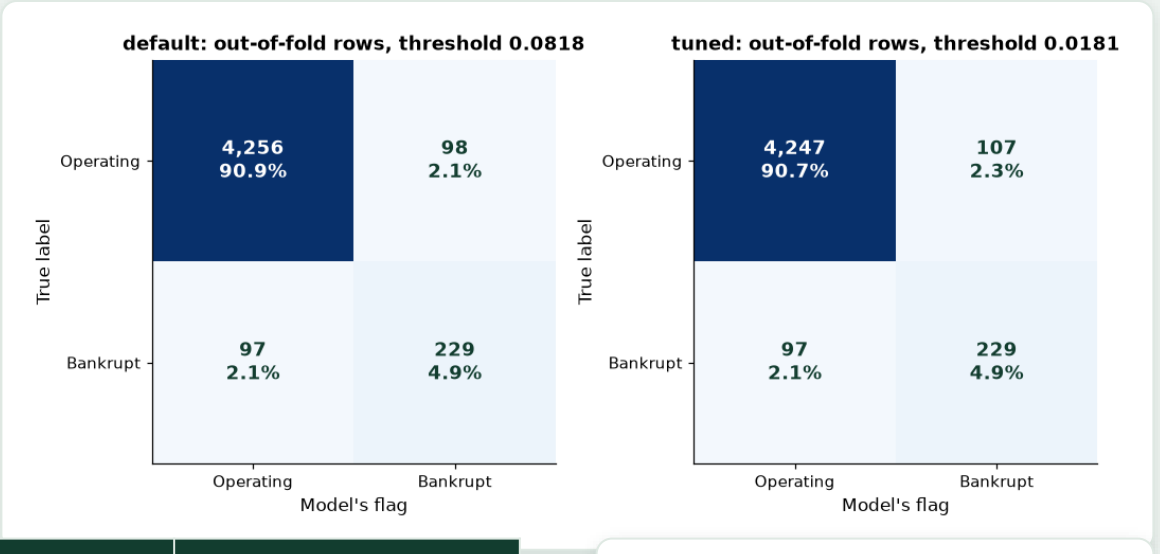
Recall at least 0.70 from out-of-fold probabilities, by Course 4's rule.

New settings change the scale of the outputs, so the threshold is chosen again after every search. The precision beside it is in-sample for the threshold.

Each output: the model probability of bankruptcy in this file. A flag is not a credit view.

Where the outputs fall, and the flags





Model	Threshold	Flagged	Recall	Precision
default	0.0818	327	0.7025	0.7003
tuned	0.0181	336	0.7025	0.6815

Median operating output: 1.03e-04 default, 1.39e-06 tuned

Both catch 229; false flags 98 and 107. Each model needs its own threshold.

Excel side by side

Excel side by side (tested in Excel)

`=AVERAGE(B2:F2)`	a candidate's mean, filled down
`=STDEV.P(B2:F2)`	std_test_score: 0.0445 for the best
`=STDEV.S(B2:F2)`	larger: 0.0497 (divides by 4)
`=MAX(G2:G7)`	best_score_ 0.7760
`=AVERAGE(A20:E20)`	nested mean 0.7751
`=MAX(A24:A27)-MIN(A24:A27)`	fold-seed range 0.0075

No randomized search and no cross-validation in Excel. It reads a search pasted onto a sheet.

Save the day: reading a commit

```
git show --stat HEAD
```

The last commit's message, then one line per file: its name, lines changed, and a bar of + for lines added.

```
model,cv_mean,cv_sd,nested_mean
lightgbm_default,0.7783,0.0469,0.7783
lightgbm_random_search,0.7760,0.0445,0.7751
```

A tuning day's log: best_score_ and the nested mean, side by side.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows choose, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: best_score_ logged as the tuned model's honest score; the test asks for the nested mean.

Code that runs is not the same as code that is right.

What you can do now

- Rebuild Course 4's Polish rows; keep its test rows closed
- A grid costs candidates times folds
- A randomized search covers more values for the same budget
- Measure a gain against reshuffled folds
- Report the nested score beside best_score_

Week 3, Day 2 Feature engineering on financial ratios